

Capstone Project - Revenue Analysis

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https://public.tableau.com/app/profile/iordanis.kefalidis/viz/capstone_dashboard_17768511626300/RevenueDashboard?publish=yes

https://github.com/JordanKef/capstone_project

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Questions/Answers and Metabase Visualisation

Question 1 :What's the % diff between May 22' and June 22'?

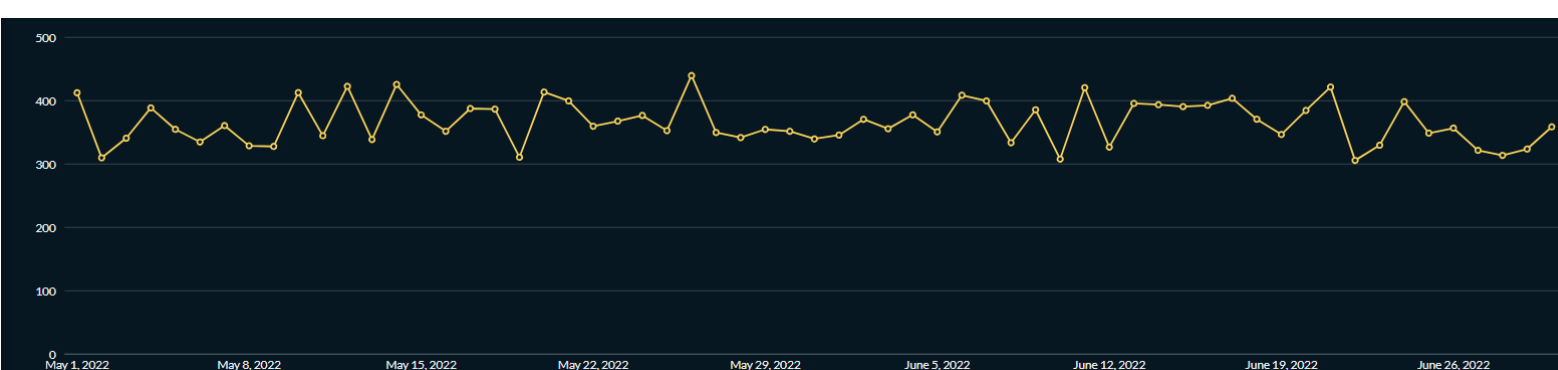
Answer : The total revenue for May 2022 was **11,347**, while for June 2022 it was **10,923**, indicating a decrease in revenue.

The percentage difference between May 2022 and June 2022 is:
 $((\text{Rev_June} - \text{Rev_May}) / \text{Rev_May}) * 100 = -3.74\%$

This indicates a decrease in revenue from May to June.

```
1 SELECT
2   date,
3   SUM(total_revenue) AS revenue
4 FROM `Reporting_DB_CapProj.rep_revenue_per_period`
5 WHERE 1=1
6 [[ AND date >= {{start_date}} ]]
7 [[ AND date <= {{end_date}} ]]
8 GROUP BY date
9 ORDER BY date
```

Start Date: May 1, 2022 x End Date: June 30, 2022 x



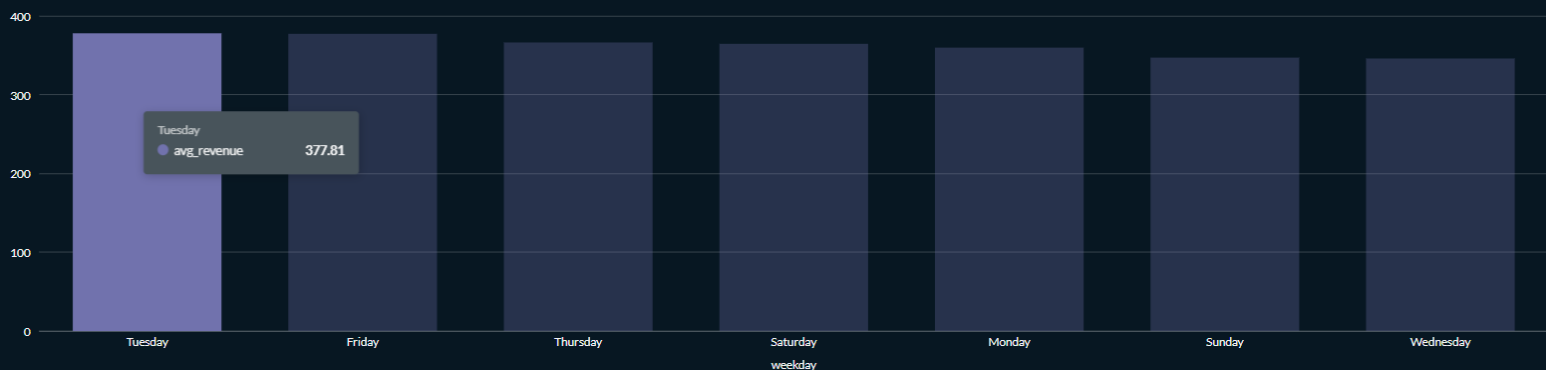
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## Question 2 :

What's the weekday with the highest average revenue?

Answer : Tuesday has the highest average revenue (377,81), followed by Friday (377,11).

```
1 SELECT
2   weekday,
3   AVG(total_revenue) AS avg_revenue
4 FROM `Reporting_DB_CapProj.rep_revenue_per_period`
5
6 GROUP BY weekday
7 ORDER BY avg_revenue DESC
```



## Question 3 : Which are the top 5 customers by revenue for June 2022 ?

```
1 SELECT customer_id, first_name, last_name, SUM(total_revenue) AS revenue
2 FROM `Reporting_DB_CapProj.rep_revenue_per_customer_and_period`
3 WHERE 1=1
4 [[ AND date >= {{start_date}} ]]
5 [[ AND date <= {{end_date}} ]]
6 GROUP BY customer_id, first_name, last_name
7 ORDER BY revenue DESC
```

Start Date: June 1, 2022

End Date: June 30, 2022

Top 5 customers by revenue for June 2022 are:

1. Dave Gardiner (ID: 569) – 58.88
2. Jeanne Lawson (ID: 200) – 56.91
3. Rosemary Schmidt (ID: 204) – 55.89
4. Marilyn Ross (ID: 80) – 51.91
5. Marion Snyder (ID: 178) – 50.91

| customer_id | first_name | last_name | revenue |
|-------------|------------|-----------|---------|
| 569         | DAVE       | GARDINER  | 58.88   |
| 200         | JEANNE     | LAWSON    | 56.91   |
| 204         | ROSEMARY   | SCHMIDT   | 55.89   |
| 80          | MARILYN    | ROSS      | 51.91   |
| 178         | MARION     | SNYDER    | 50.91   |

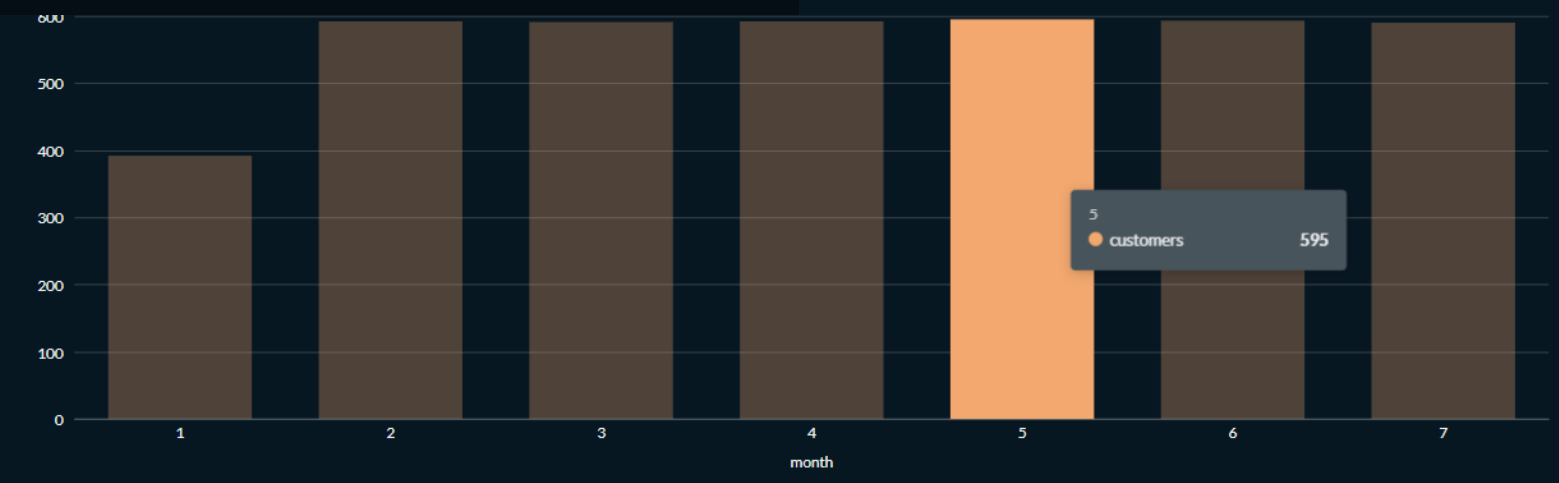
## Question 4 : Which is the month with the most unique customers?

Answer : May has the highest number of unique customers (595), followed by June (593).

```
1 SELECT
2   month,
3   COUNT(DISTINCT customer_id) AS customers
4 FROM `Reporting_DB_CapProj.rep_revenue_per_customer_and_period`
5 WHERE total_revenue > 0
6 [[AND date BETWEEN {{start_date}} AND {{end_date}}]]
7 GROUP BY month
8 ORDER BY customers DESC
```

📅 Start Date: January 1, 2022 ✕

📅 End Date: January 1, 2023 ✕

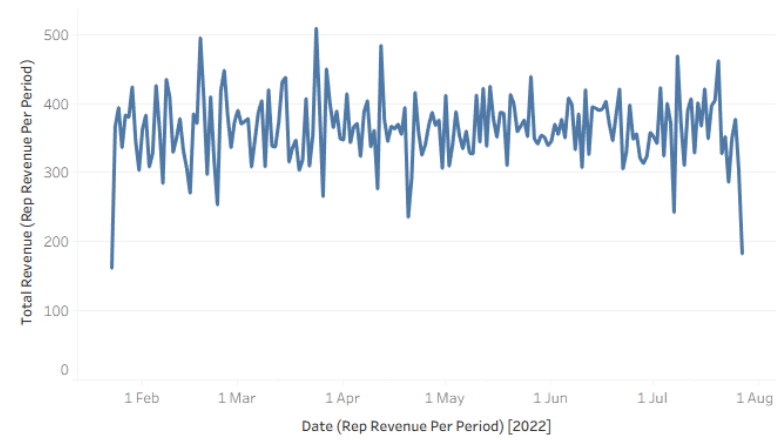


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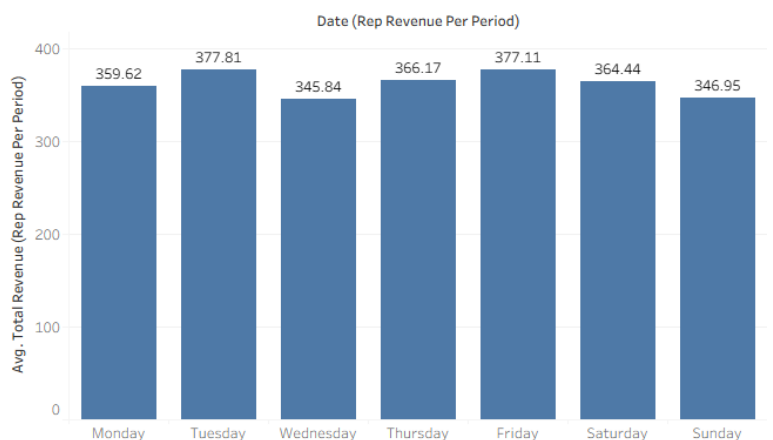
Dashboard

Revenue Analysis Dashboard

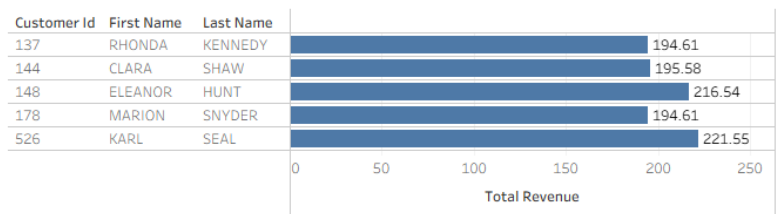
Revenue Over Time



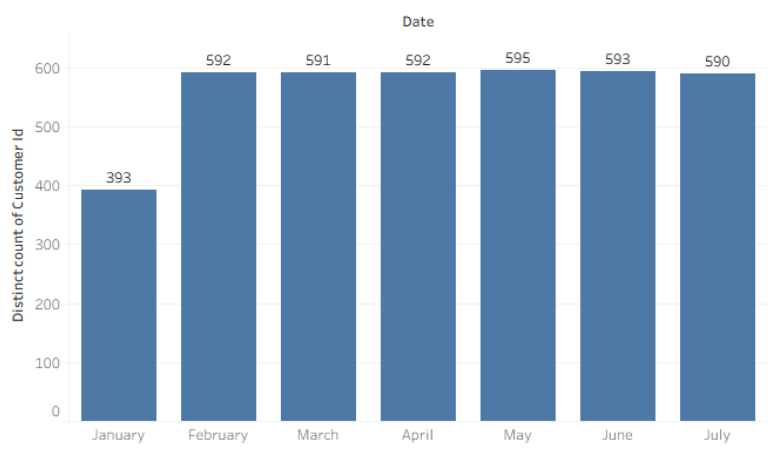
Revenue by Weekday



Top Customers



Customers per Month



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Methodology & Approach

This project was developed following a structured workflow that I learned during the course.

At the beginning, raw data was loaded into staging tables (stg_*) in order to keep a clear separation between raw and transformed data.

Then, I created reporting tables:

- rep_revenue_per_period
- rep_revenue_per_customer_and_period

to make the data easier to use for analysis and visualization.

To support time-based analysis, I also created a simple date spine (all_dates) and a reporting periods table. This helped ensure that calculations work correctly across different time ranges.

The final data (and queries) was used both in Tableau (for dashboards) and in Metabase, where I implemented SQL queries with filters.

ETL Process (Python / Jupyter)

As part of the project, I used Python notebooks (Jupyter) to perform the ETL process.

In this step:

- I extracted data from the source database
- Transformed the data into staging tables (stg_*)
- Created the final reporting tables (rep_*)

This helped automate the data preparation process and better understand how data pipelines work in practice.

Testing & Validation

During development, I tested my queries step by step to make sure everything works correctly.

Some of the checks I performed:

- Comparing results between staging and reporting tables
- Verifying totals (e.g., revenue, customers)
- Checking that Tableau and SQL queries return the same results

This process helped me identify and fix issues early.

Data Pipeline Overview

The project is structured in three main layers:

1. **Staging Layer (stg_*)**
Raw data with minimal changes
2. **Reporting Layer (rep_*)**
Aggregated data ready for analysis
3. **BI Layer (Tableau & Metabase)**
Visualizations and insights

Tools & Technologies

- BigQuery (Data Warehouse for data storage and querying)
- PostgreSQL (Source system with Pagila Database)
- Python (Jupyter Notebooks for ETL steps)
- Tableau Public (Dashboard)
- Metabase (SQL queries & filters)
- GitHub (version control to manage the project files)